Semiparametric bivariate modelling with flexible extremal dependence

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Joint work with Dani Gamerman (UFRJ)

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SCHOOL OF
HUMAN SCIENCES
& TECHNOLOGY

Goal, Introduction, and Motivation

Precise knowledge and predicting capabilities for extremes are fundamental in many disciplines:

- Environmental sciences
- Finance and actuarial science
- Engineering and reliability

Evidence of increasing occurence of extremes and larger insurance and economic losses.

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Assessment of extreme dependence is often critical':

- sea level and wave height
- concentration of O_3 and NO_2

Extreme value theory provides the theoretical framework to perform inference over extremes.

Univariate EVT

The peaks over threshold approach states that the only possible non-degenerate limiting distribution of properly rescaled exceedances over a threshold u is the *generalized Pareto distribution* (GPD) with CDF

$$H_{\xi,\sigma,u}(x) = \begin{cases} 1-(1+\xi\frac{x-u}{\sigma})^{-1/\xi}, & \text{if } \xi \neq 0 \\ 1-\exp(-\frac{x-u}{\sigma}), & \text{if } \xi = 0 \end{cases}$$

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Drawbacks for inference:

- Small number of observations retained (bigger variance);
- Choice of threshold u critical.

Extreme Value Mixture Models

For this reason, various approaches using the full dataset have been proposed. In general

$$f_{\boldsymbol{\theta}}(x) = c_{\boldsymbol{\theta}}((1 - p_{u,\tau}(x))g_{\boldsymbol{\gamma}}(x) + p_{u,\tau}(x)h_{\xi,\sigma,u}(x))$$

- $h_{\xi,\sigma,u}$ is the GPD density
- $g_{\boldsymbol{\gamma}}$ is a flexibile distribution for the bulk of the data
- $p_{u,\tau}$ is a weight function
- $c_{\boldsymbol{\theta}}$ is a normalizing constant

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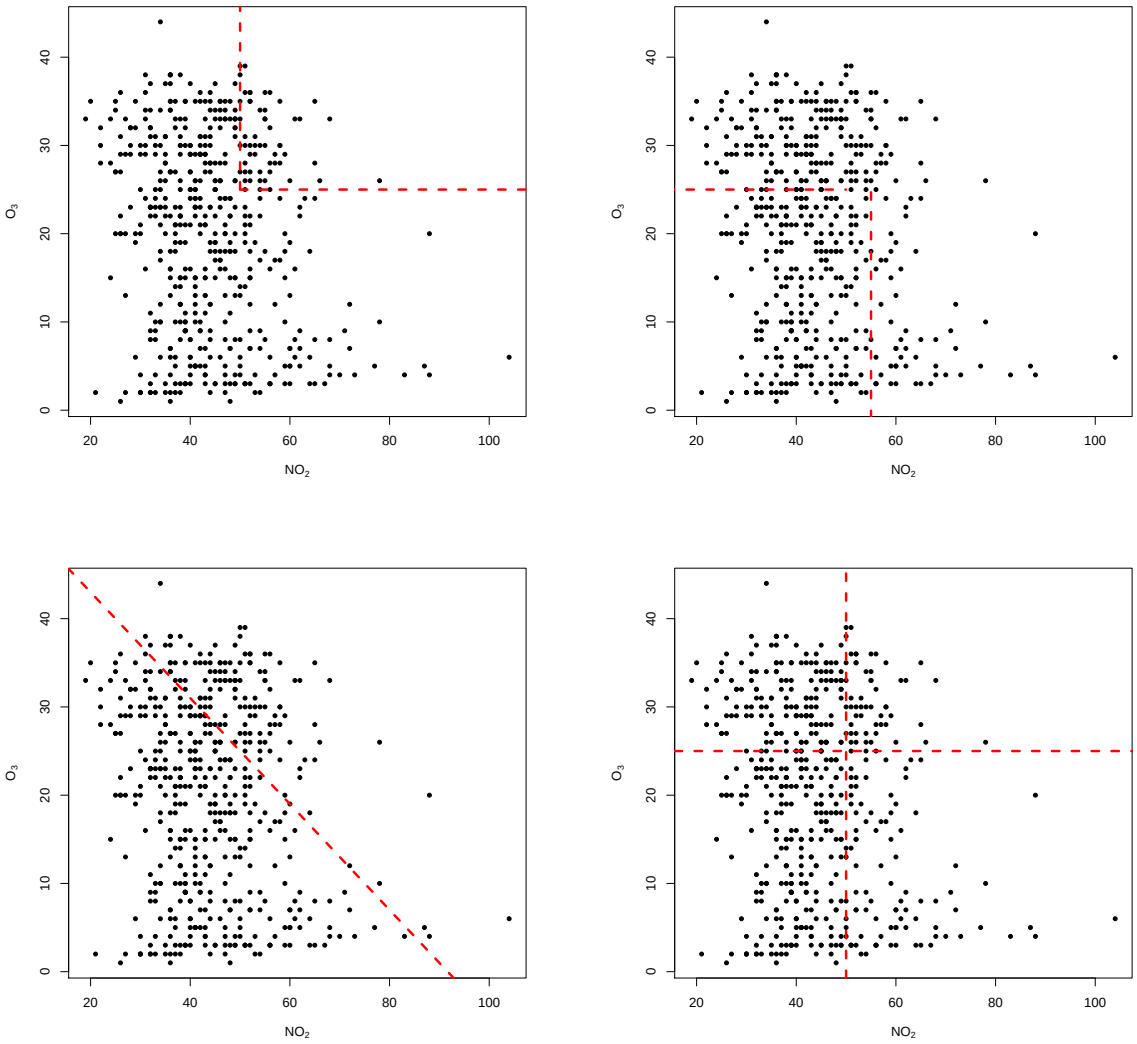
Often $p_{u,\tau}$ is chosen such that

$$f_{\boldsymbol{\theta}}(x) = \begin{cases} g_{\boldsymbol{\gamma}}(x), & \text{if } x \leq u \\ (1 - G_{\boldsymbol{\gamma}}(u))h_{\xi,\sigma,u}(x) & \text{if } x > u \end{cases}$$

Multivariate Extremes

- Asymptotic results about the distribution of extremes in random vectors exist;
- The resulting distribution does not have a closed parametric form as in the univariate case;
- For this reason, a variety of approaches have been proposed in the literature, either parametric, non-parametric or based on different theoretical justifications;
- It is customary to focus only on the dependence structure and perform inference in a 2-steps approach

Threshold Selection



Extremal Dependence

Consider two random variables X_1 and X_2 with CDFs F_1 and F_2.

Define

$$\bar{\chi} = \lim_{u \to 1} P(F_1(X_1) > u | F_2(X_2) > u)$$

This is usually called the coefficient of asymptotic dependence.

- if $\bar{\chi} = 0 \Longrightarrow$ asymptotic independence
- if $\bar{\chi} \in (0, 1] \Longrightarrow$ asymptotic dependence

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Although recently there have been efforts to define models that encompass both independence and dependence, traditionally models could only consider one of the two (most often dependence).

Our Modelling Approach

We wanted to achieve the following objectives:

- Use the full dataset to investigate extremal dependence;
- Encompass both types of extremal dependence;
- Model contemporaneously marginals and dependence.

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A starting point is the model

$$f_{\boldsymbol{\theta}}(x_1, x_2) = (1-p) f_{\theta_{ind}}(x_1, x_2) + p f_{\theta_{dep}}(x_1, x_2)$$

where

- $f_{\theta_{ind}}$ is an asymptotically independent model
- $f_{\theta_{dep}}$ is an asymptotically dependent model

Our Modelling Approach

To split the task of choosing a model for marginals and one for dependence, we used copulae

$$F(x_1, x_2) = C(F_1(x_1), F_2(x_2)),$$

where

- F_1 is the MGPD model of Nascimento et al. (2012)
- C is a copula function modelling dependence
  - F is asymptotically dependent if C is chosen as a mixture of T or skew-T copulae;
  - F is asymptotically independent if C is chosen as a mixture of Normal or skew-Normal copulae.

Our Modelling Approach

For the model

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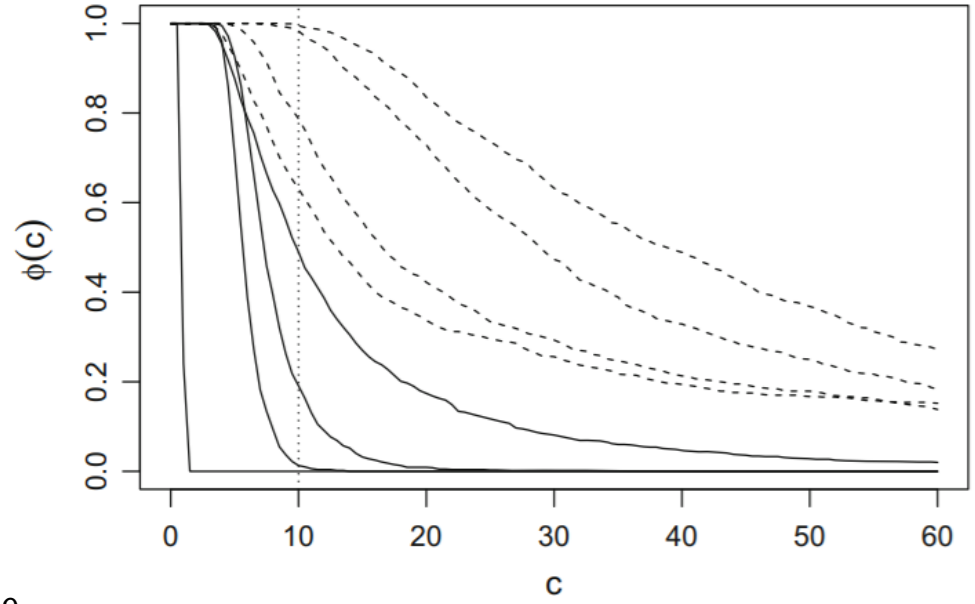
- The T-copulae were dominating the estimation process;
- The degrees of freedom of the T-copulae were the parameter characterizing the strength of extreme dependence.

The ϕ Criterion

Let v be the degrees of freedom of a T-copula and define

$$\phi(c) = P(v \in [c, +\infty))$$

- Values of $\phi(c)$ close to zero give an indication of extreme dependence
- Values of $\phi(c)$ close to one give an indication of extreme independence
- Of course, the values depend on the choice of c



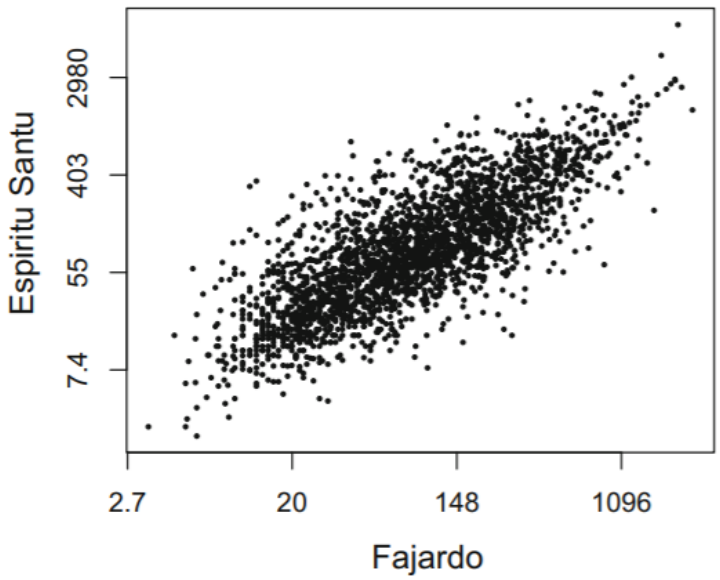
Estimation and Implementation

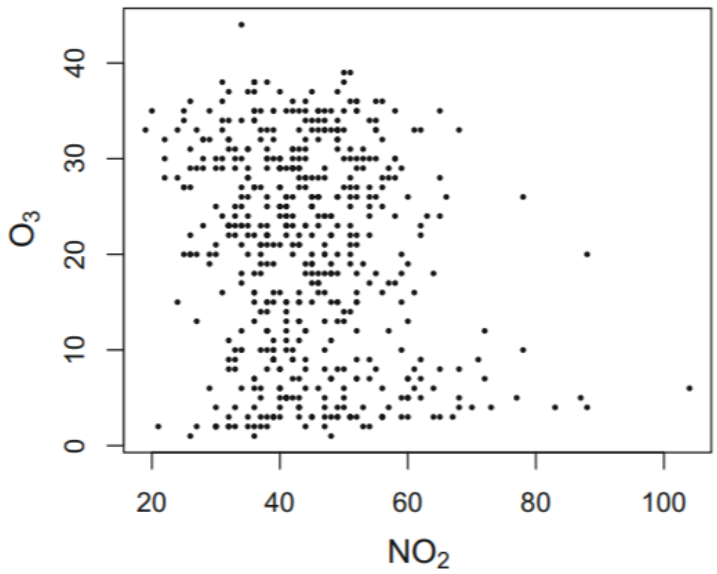
- Inference is carried out in a Bayesian fashion using approximating MCMC algorithms with Metropolis-Hastings steps;
- Vague priors are defined for the parameters;
- Algorithms implemented in OX;
- Variances of the proposals tuned via adaptive Metropolis-Hastings;
- The Bayesian approach allowed us to easily compute estimates and credibility intervals for various functions of interest (e.g. quantiles, probability of joint exceedance, predictions etc.)

Data Analysis

We analyzied two datasets:

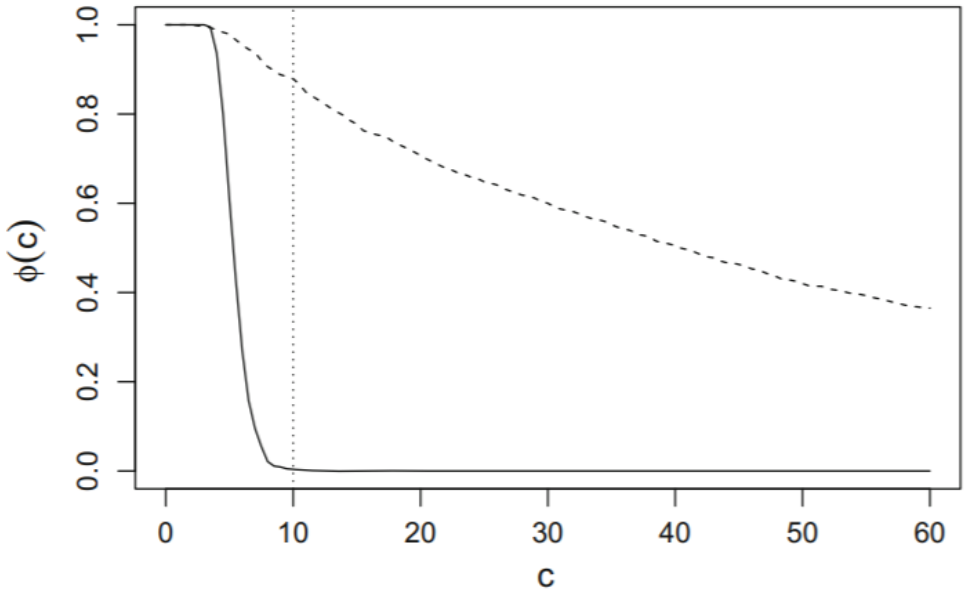
- weekly maxima from August 1966 to June 2016 of the flows of Fajardo and Espiritu Santu rivers in Puerto Rico, comprising 2492 observations (Nascimento et al. 2012);
- daily maxima of the hourly means during the winter months in 1994–1998 of NO2/O3 concentrations in Leeds, comprising 532 observations (Heffernan and Tawn 2004).



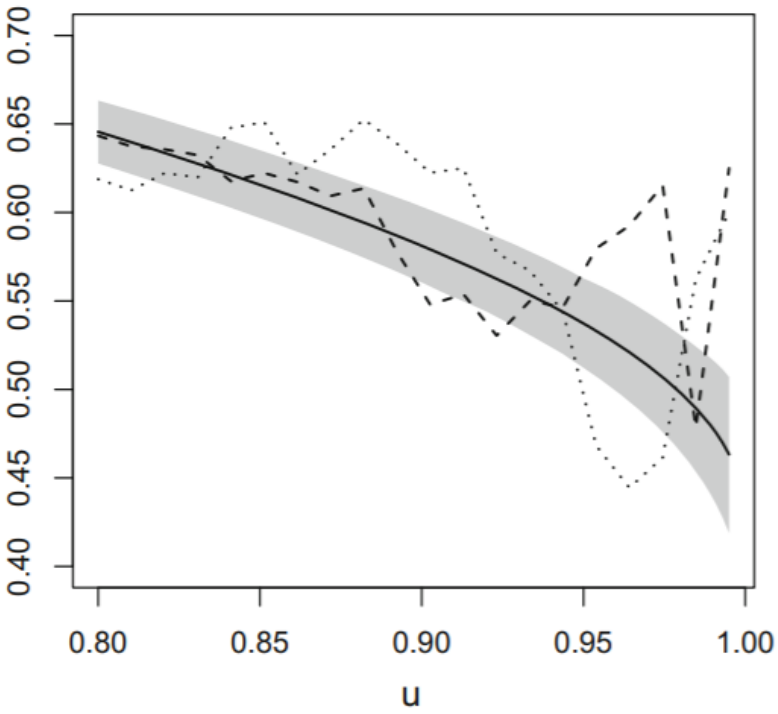


Asymptotic Dependence

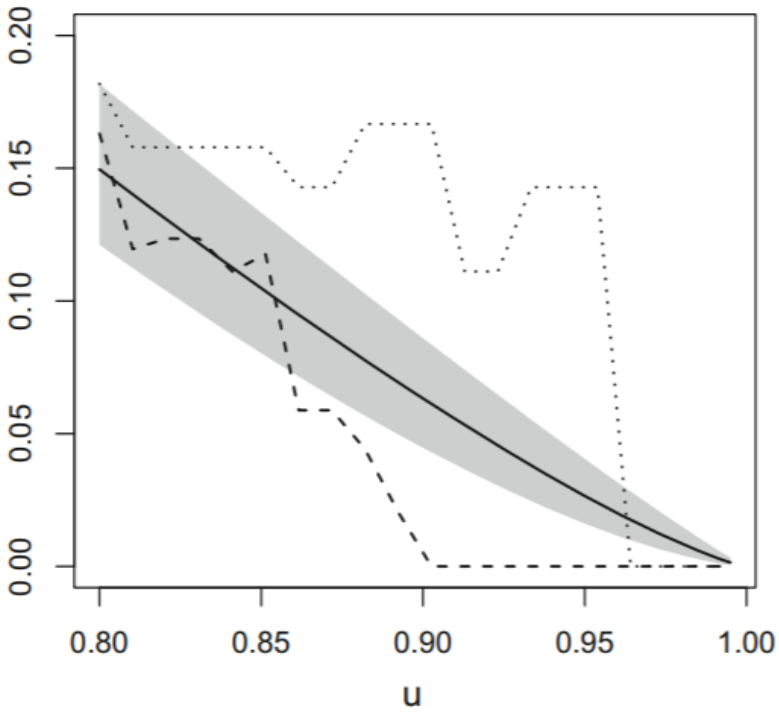
- $\phi(10) = 0.003$ for Puerto Rico rivers
- $\phi(10) = 0.97$ for Leeds Pollutants



Coefficient of Asymptotic Dependence



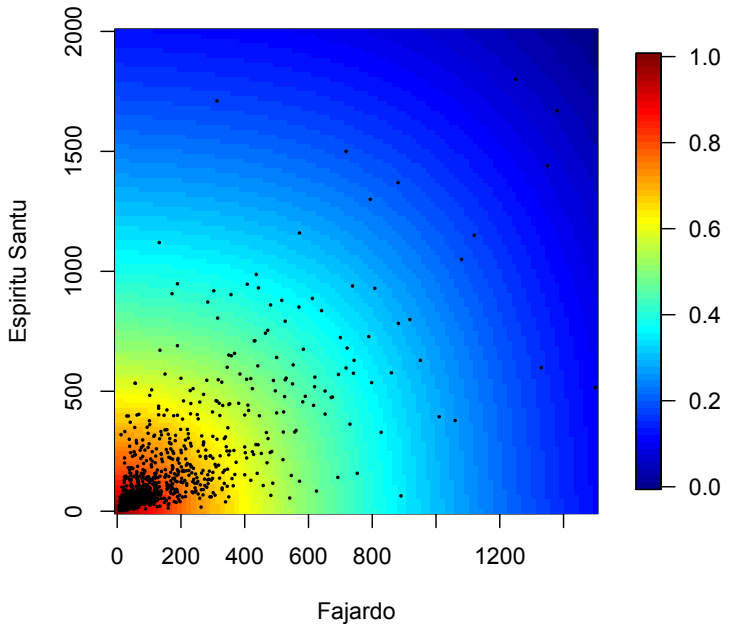
(a) $\chi(u|\Theta)$: Puerto Rico rivers

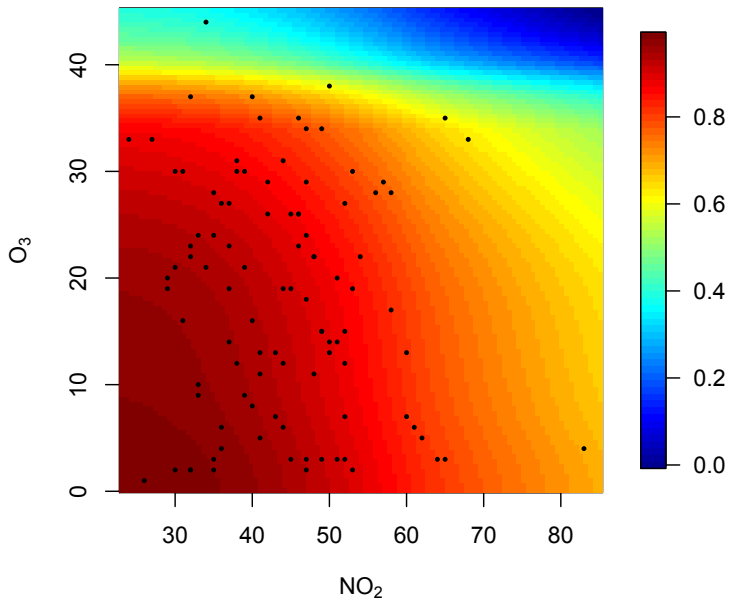


(b) $\chi(u|\Theta)$: Leeds pollutants

Additional Summaries

- Improved estimation of high quantiles compared to univariate approaches;
- Accurate estimation of the probabilities of joint exceedance;
- Predictive maps of the probabilities of joint exceedance.





Estimation over Extremes Only

In order to assess if the bulk of the data was affecting the extremes estimation process, we estimated the corresponding extreme value copulae over the extreme observations only.

	Puerto Rico rivers: χ	Leeds pollutants: $\bar{\chi}$
Full dataset	0.45 (0.39,0.50)	-0.13 (-0.21,-0.04)
Extreme points	0.43 (0.35,0.51)	-0.23 (-0.48,0.08)

Discussion

- We proposed a flexible approach to perform inference over extremal dependence not requiring the selection of an arbitrary threshold;
- In the applications considered it provided an improved estimation of various quantities of interest
- The ϕ criterion has proven to be quite robust in identifying the appropriate extremal dependence class
- An analysis over the extreme points only showed that the bulk did not bias the extremal inference.

Future Research

- Our starting model was of the form

$$f_{\boldsymbol{\theta}}(x_1, x_2) = (1-p) f_{\theta_{ind}}(x_1, x_2) + p f_{\theta_{dep}}(x_1, x_2)$$

  As in the univariate case we could select a more general weight function, $p_{x_1,x_2,\tau}$, but inference becomes challenging. Also care must be taken to properly define coherent marginal distributions.

- Multivariate vectors with different types of pairwise extreme dependence could be investigated using the theory of vine-copulae.